Title Slide

Ensemble Models vs Single Models

- How we generated evidence and analyzed it
- Classification and regression focus

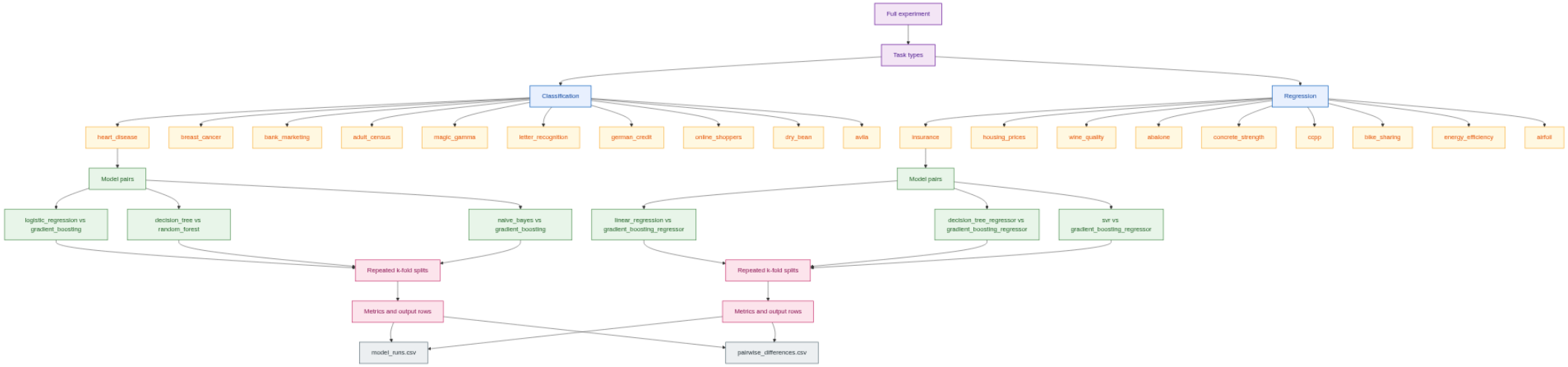
Problem Statement

- Do ensembles consistently beat single models?
- Does result change by dataset?
- Does result change by metric?

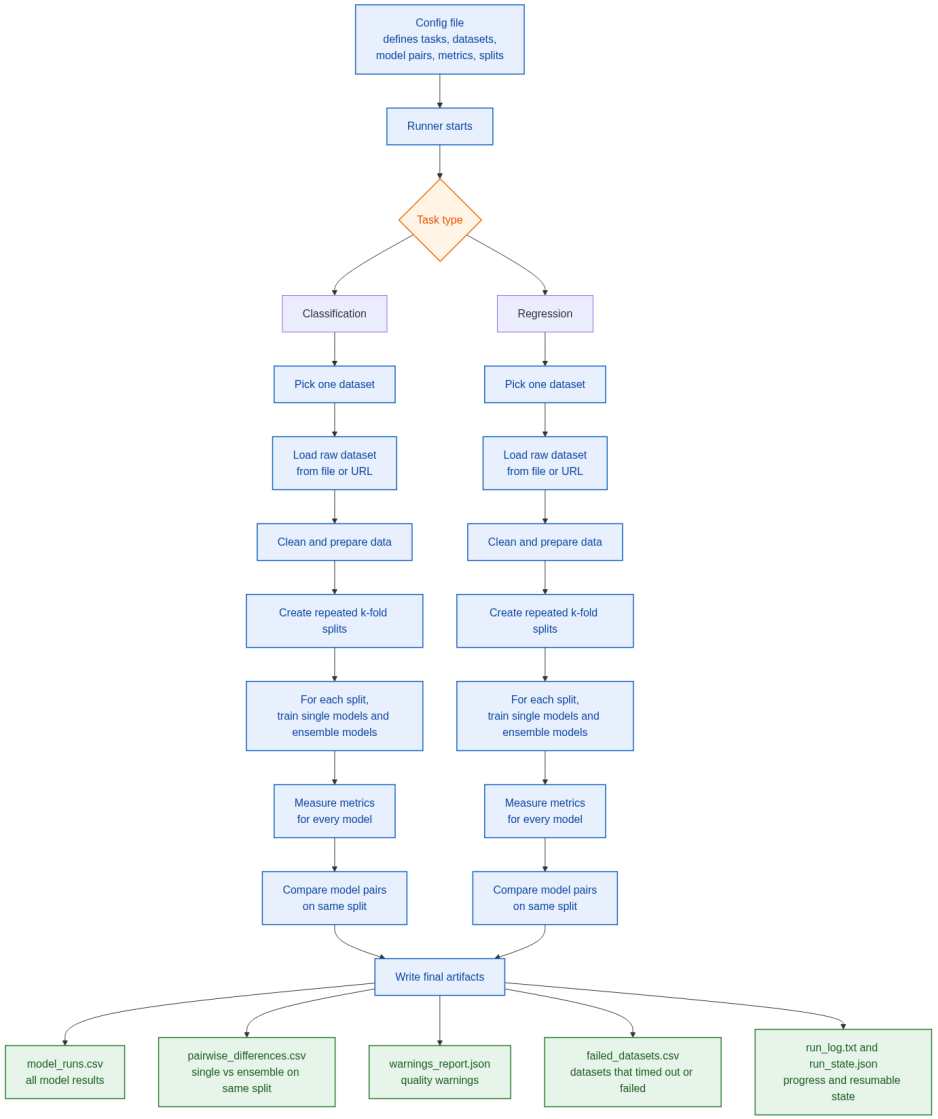
Objectives

- Build fair comparisons
- Measure paired differences
- Analyze results from multiple views
- Reach defendable conclusion

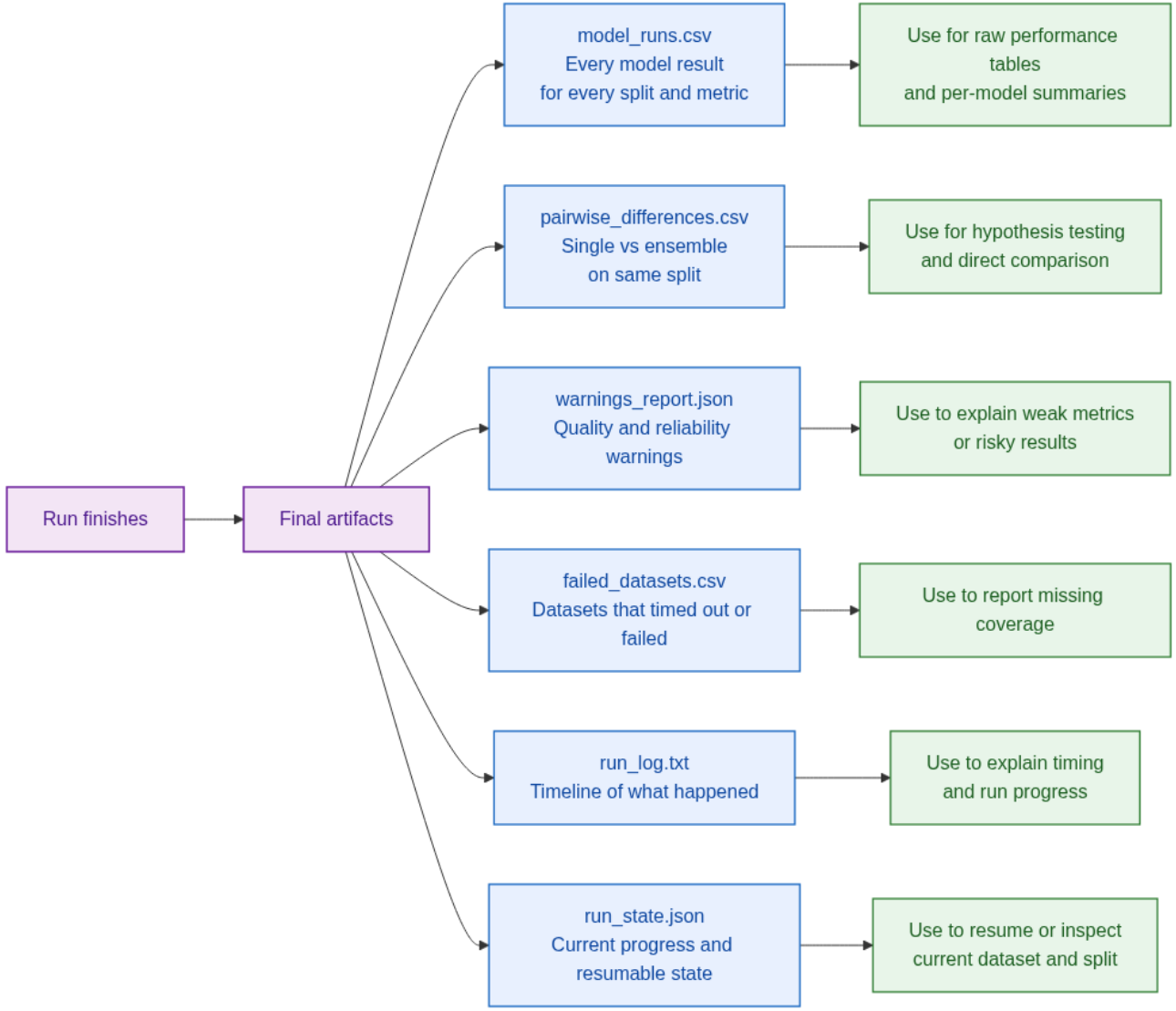
Data Collection



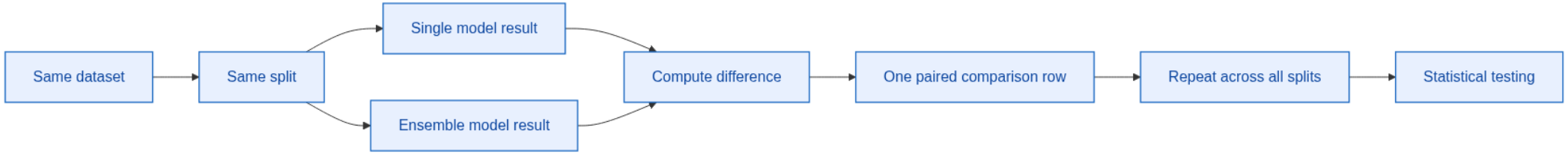
Data Collection



Data Collection

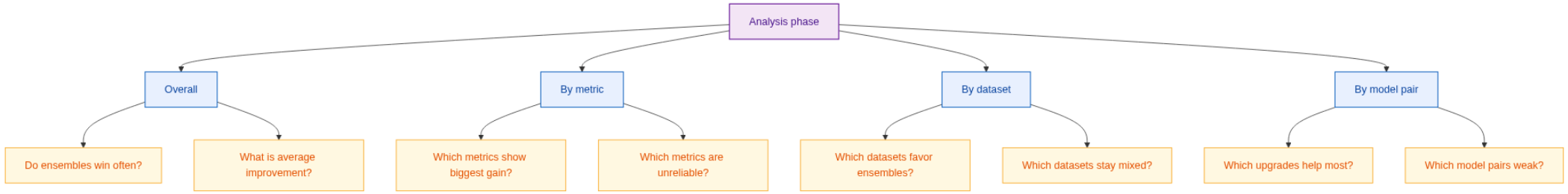


Data Preprocessing

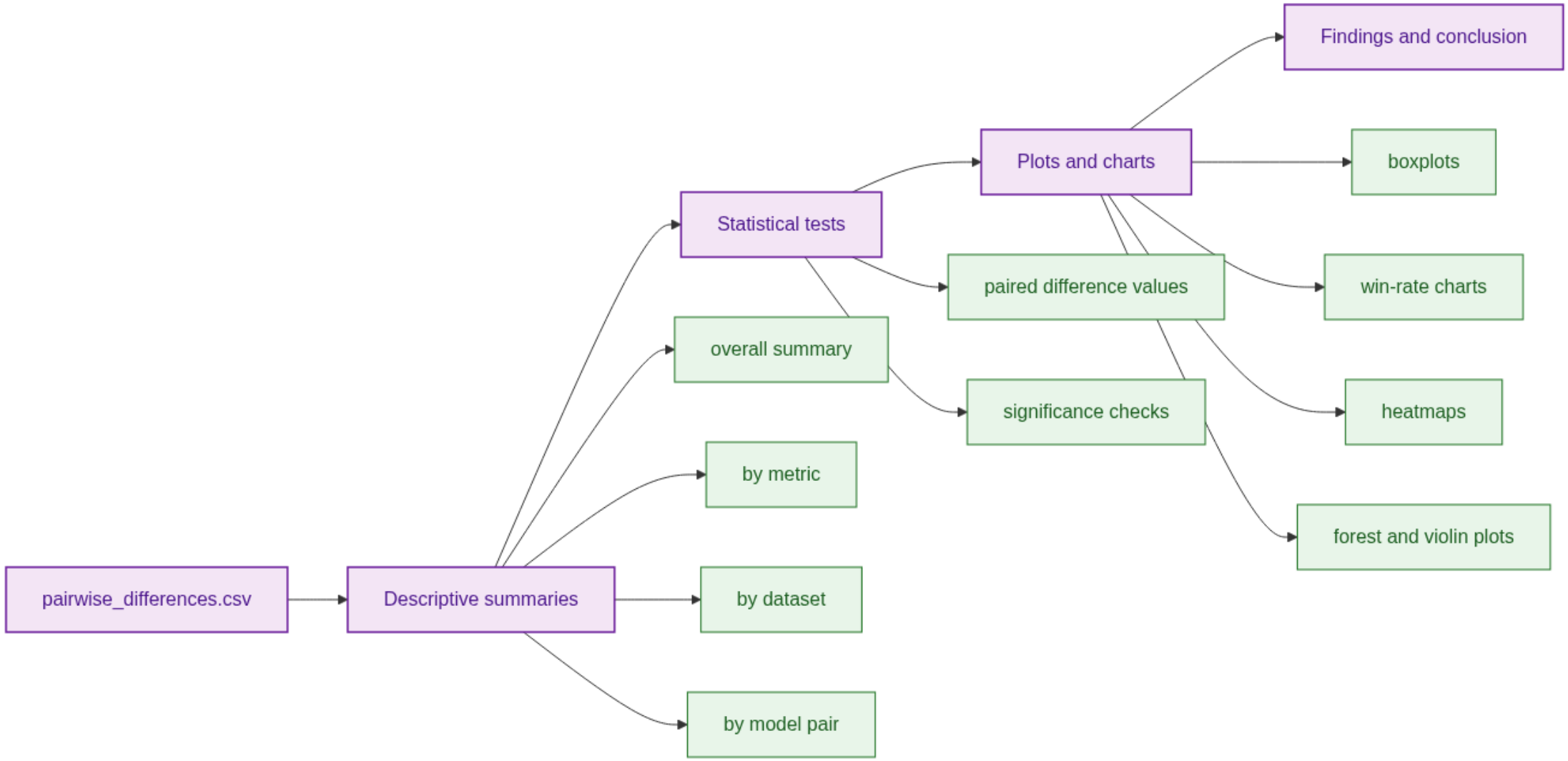


- Build matched comparisons from same split
- Convert raw outputs into analysis-ready comparison rows

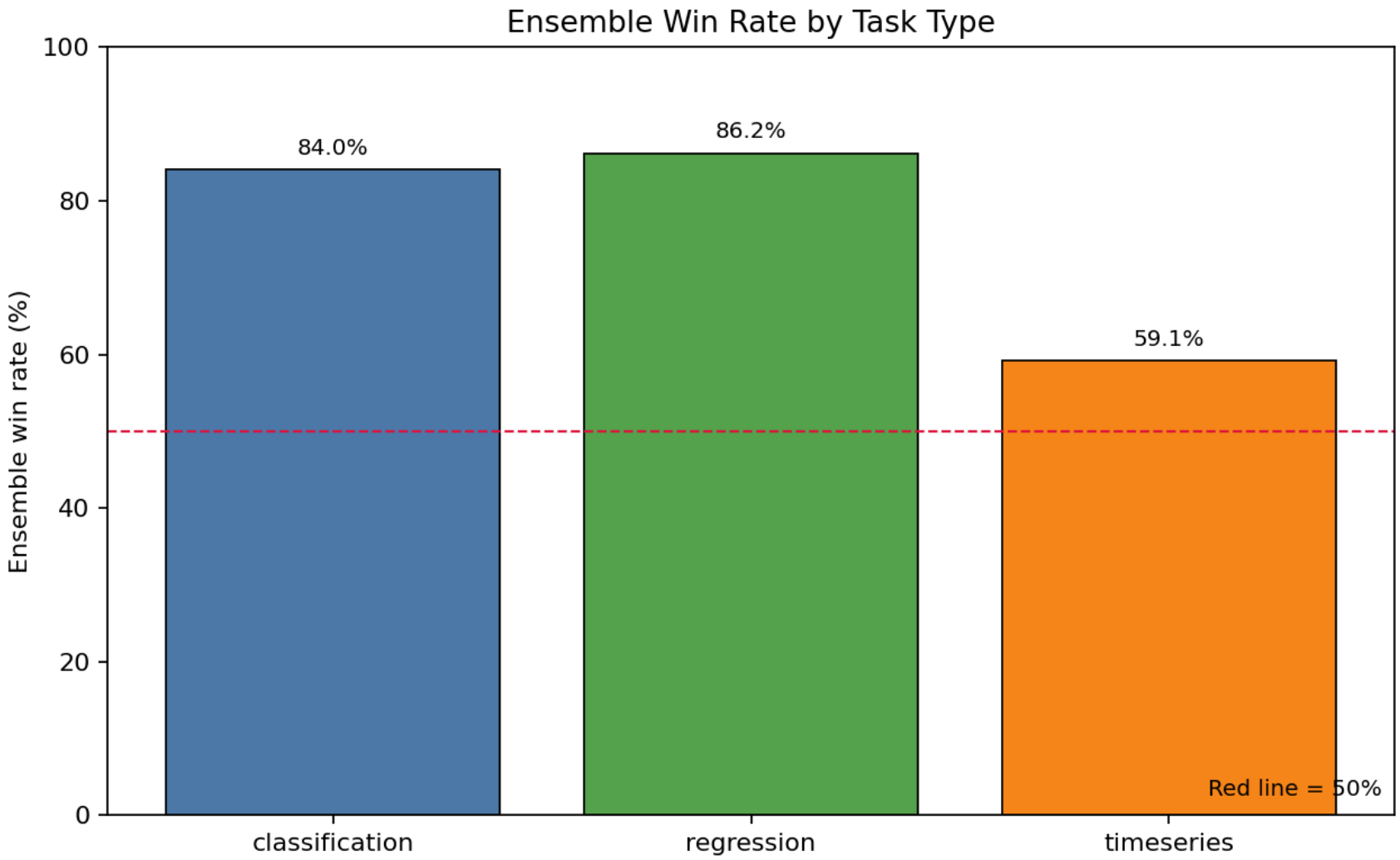
Descriptive Analysis



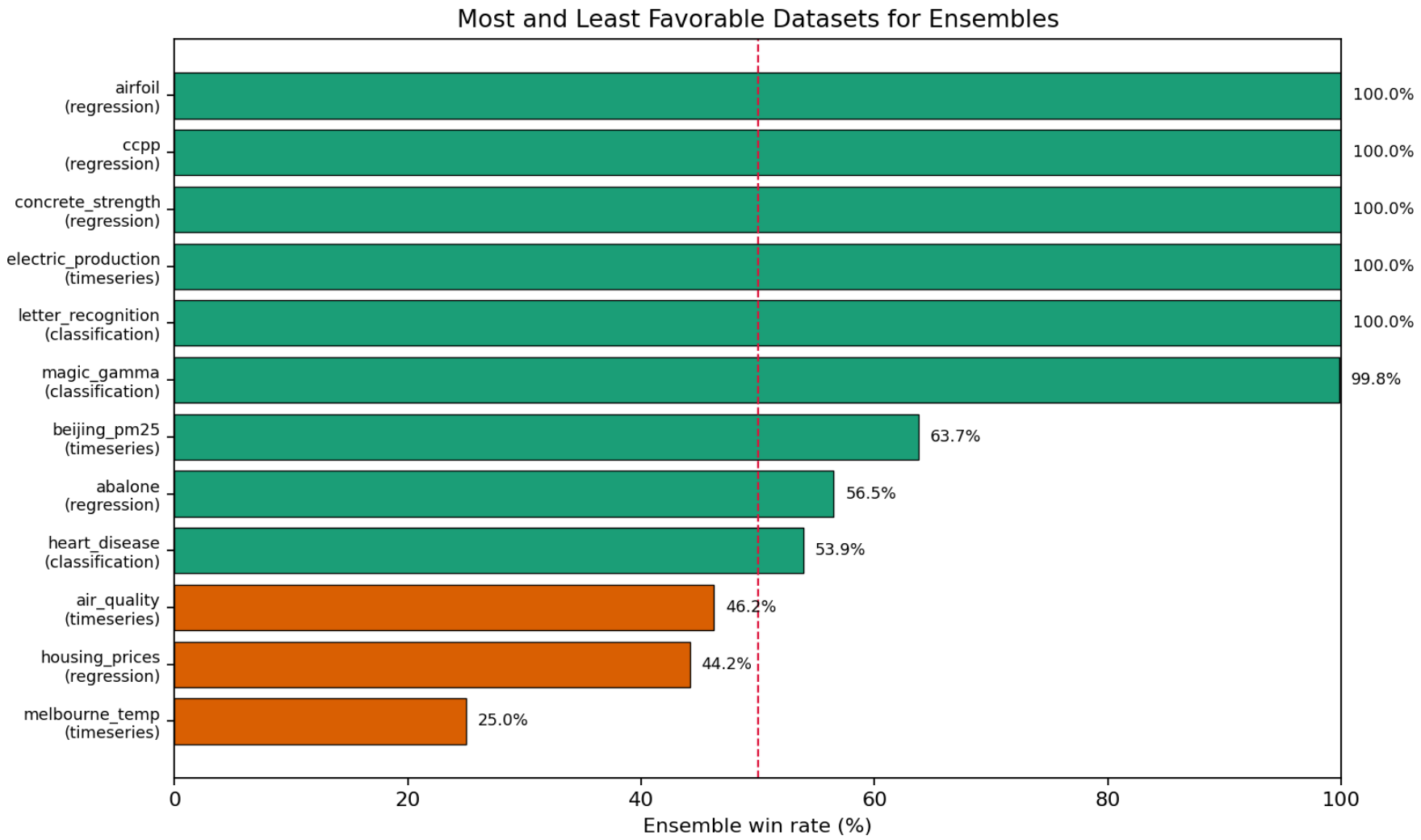
Descriptive Analysis



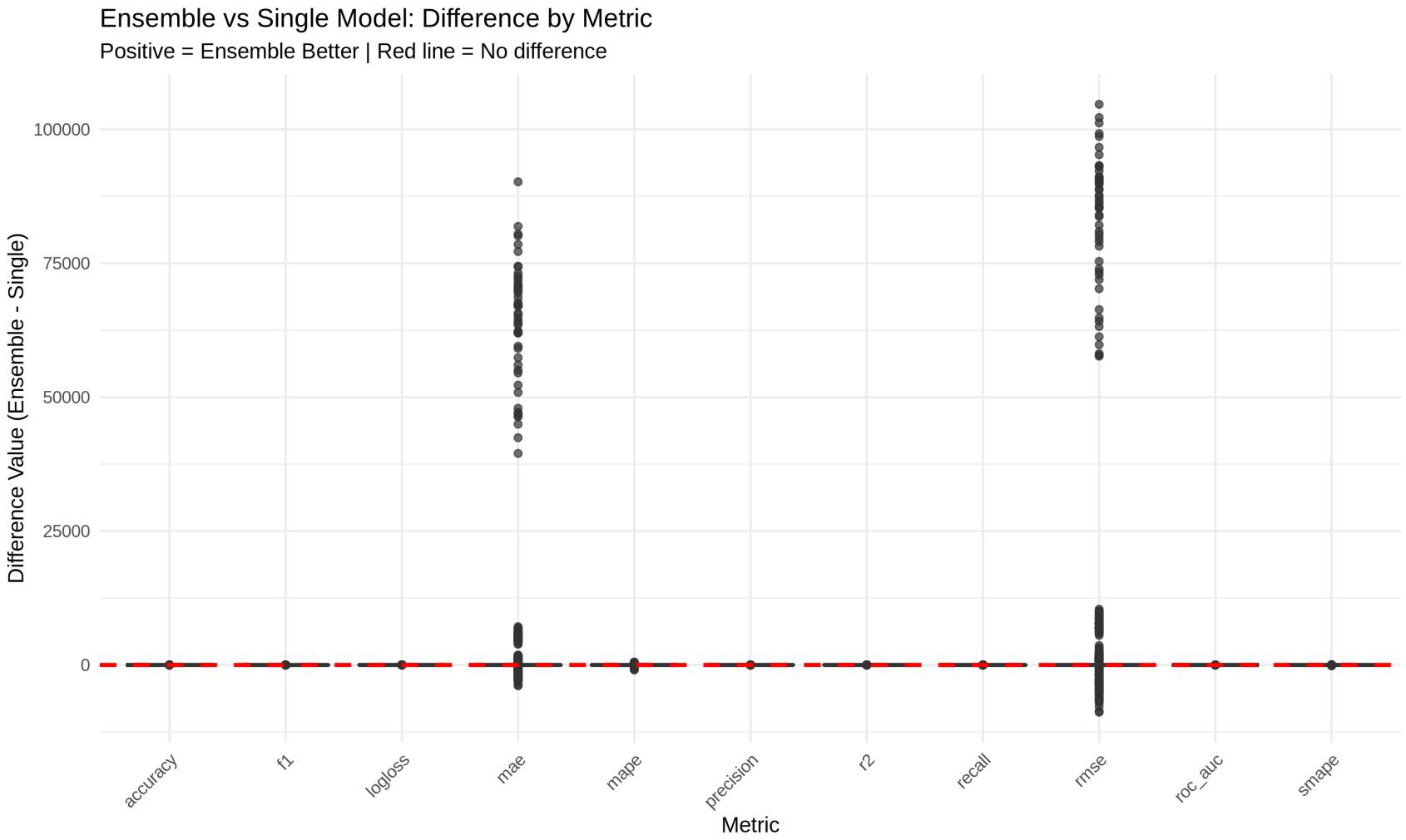
Descriptive Analysis



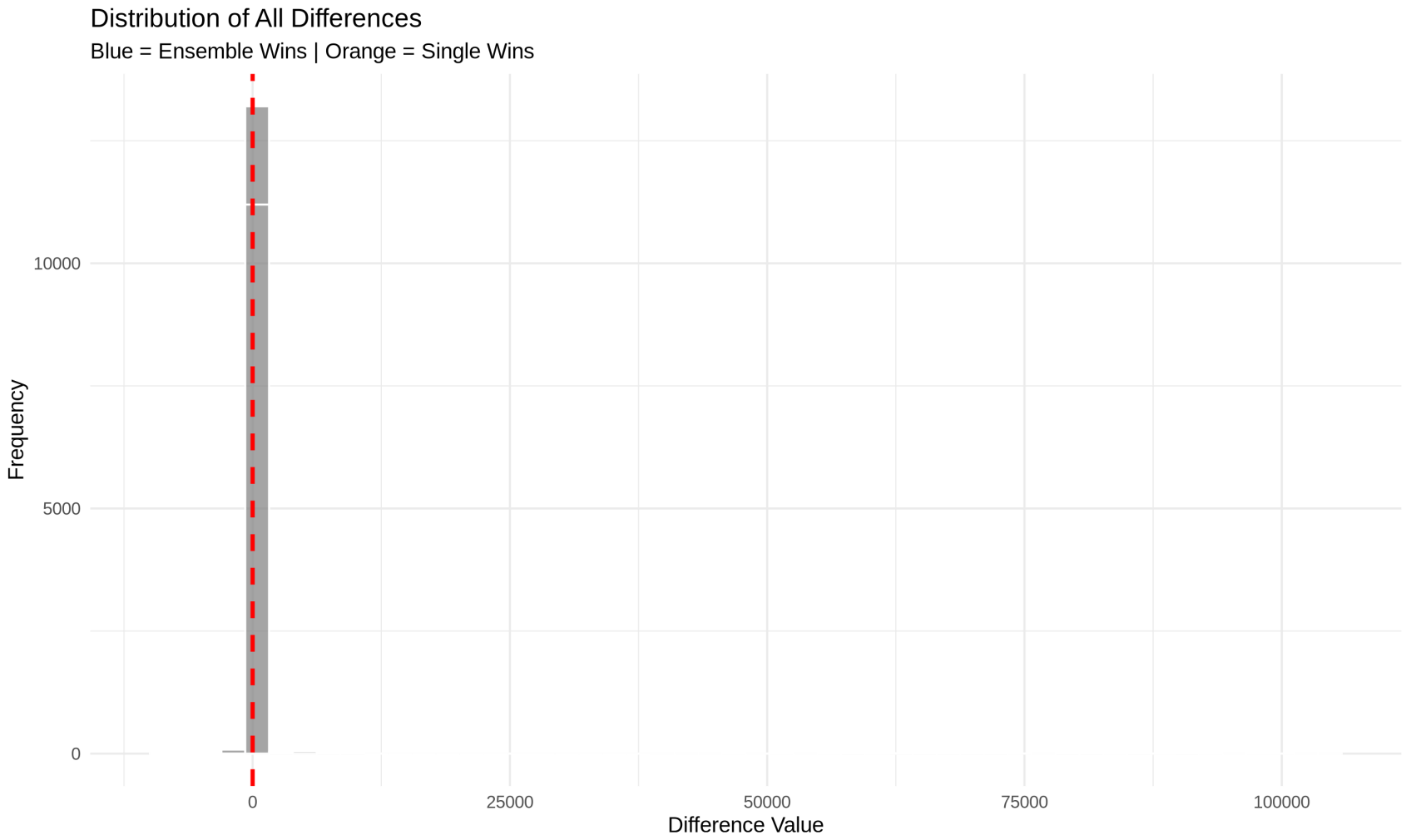
Descriptive Analysis



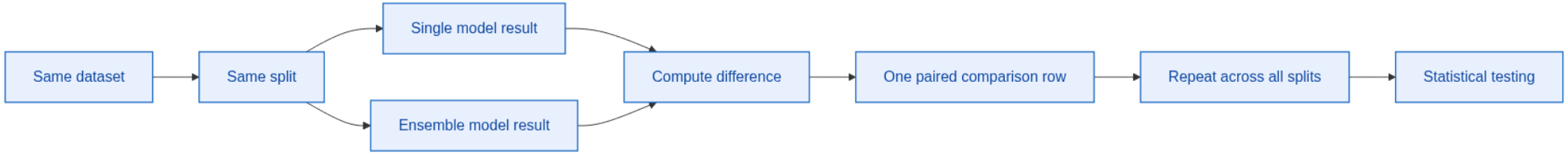
Descriptive Analysis



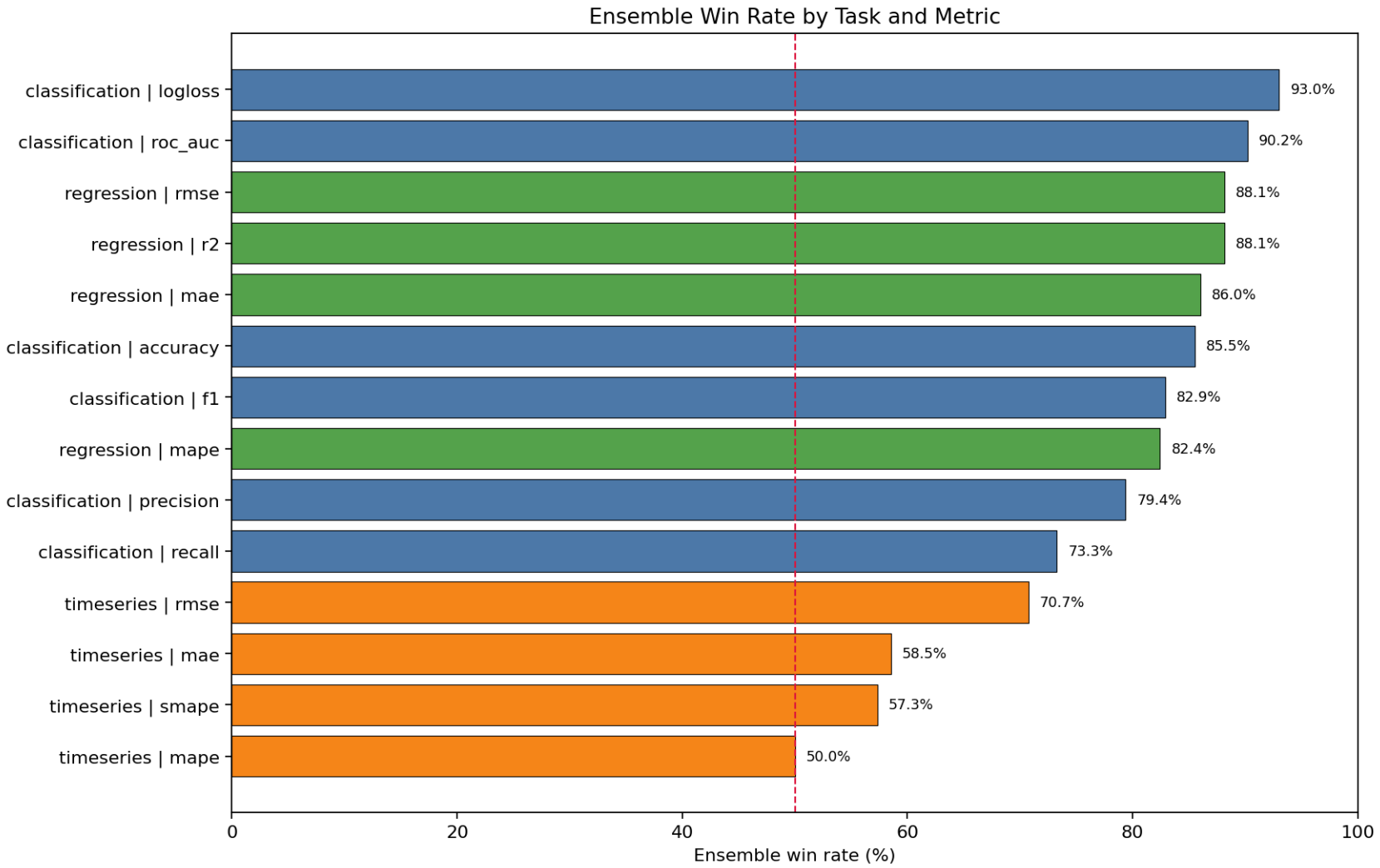
Descriptive Analysis



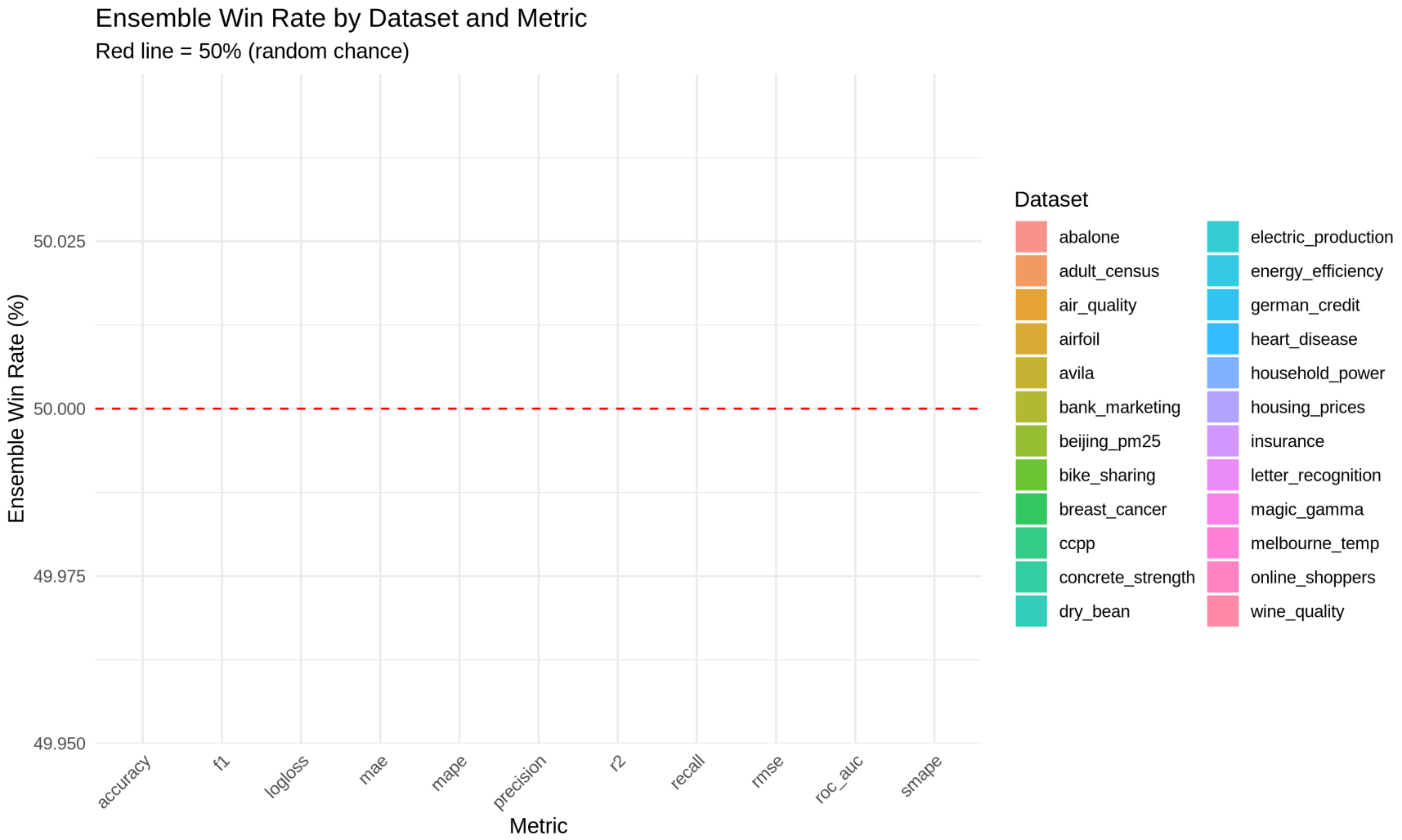
Inferential Analysis



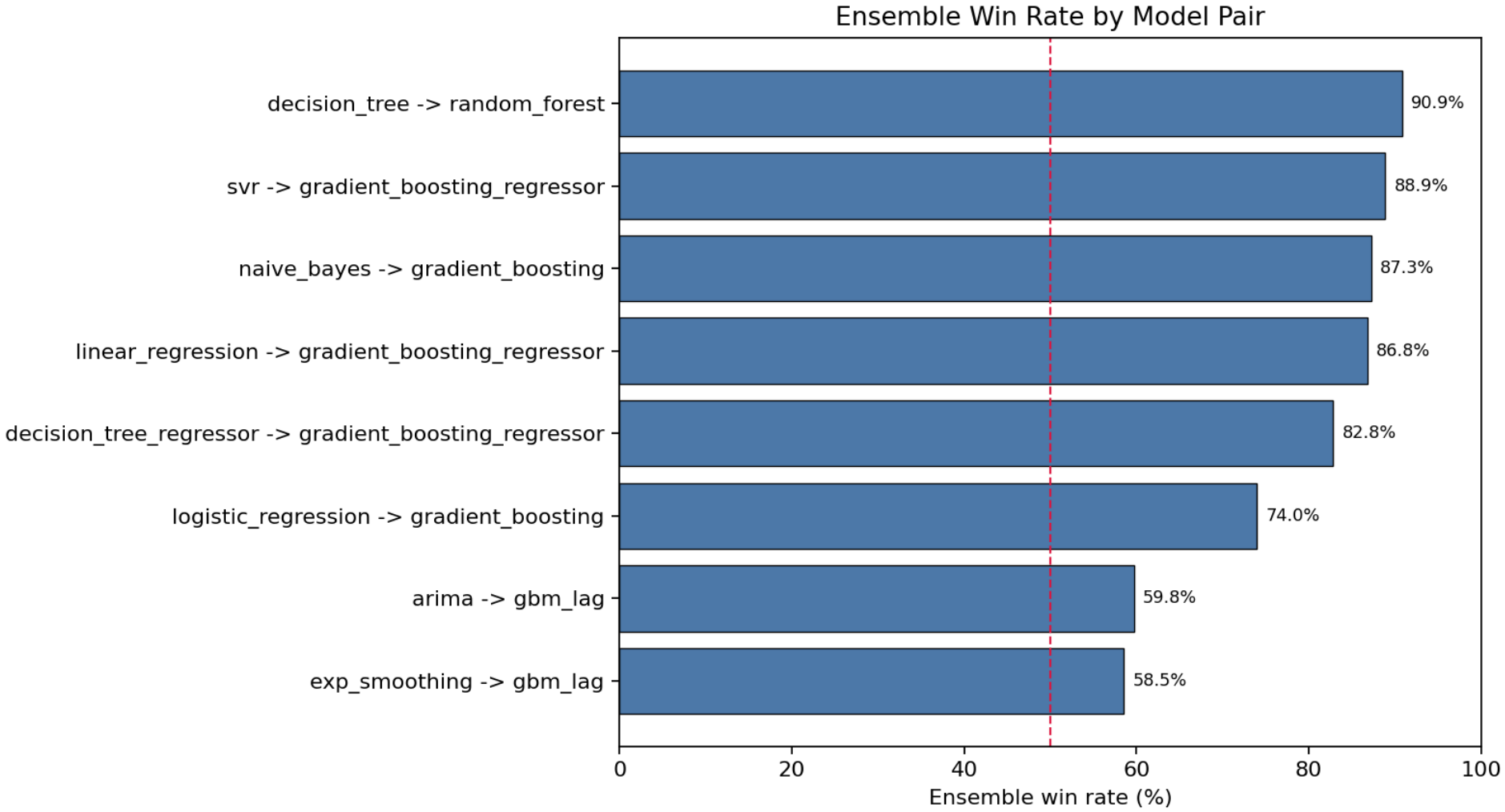
Inferential Analysis



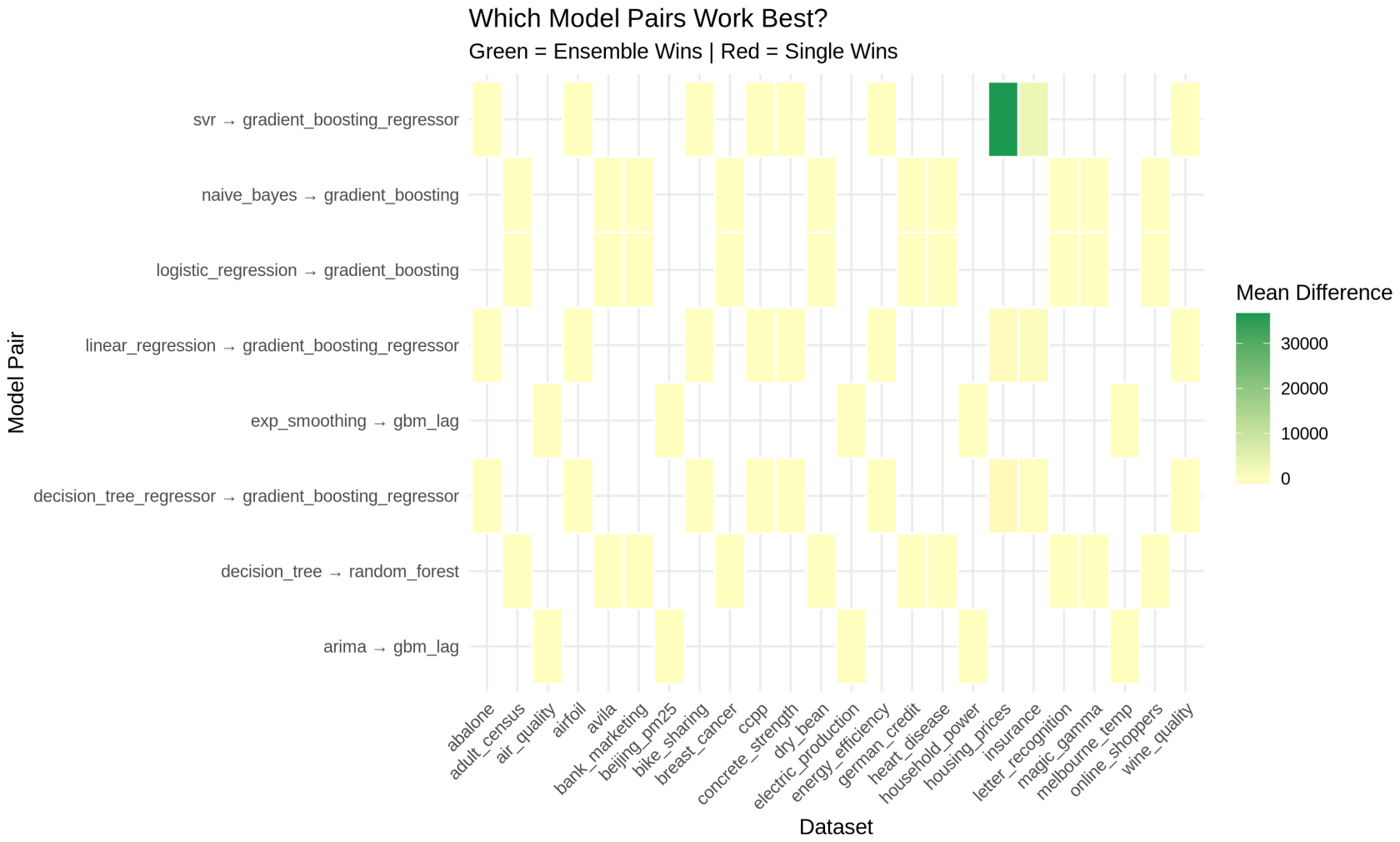
Inferential Analysis



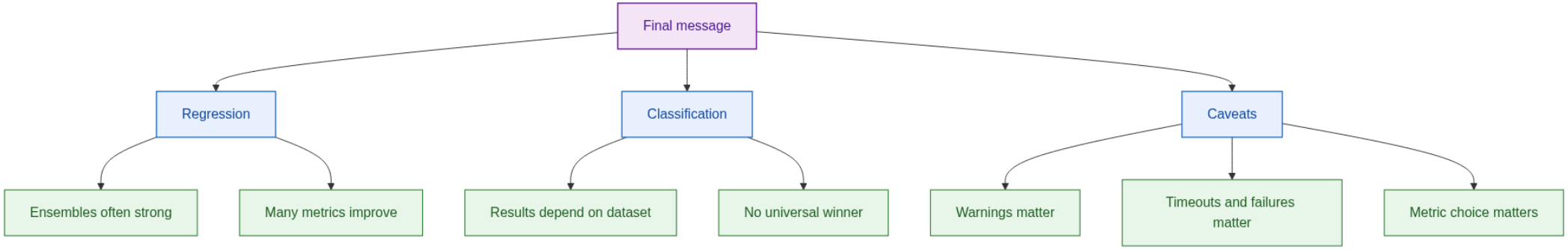
Predictive Model



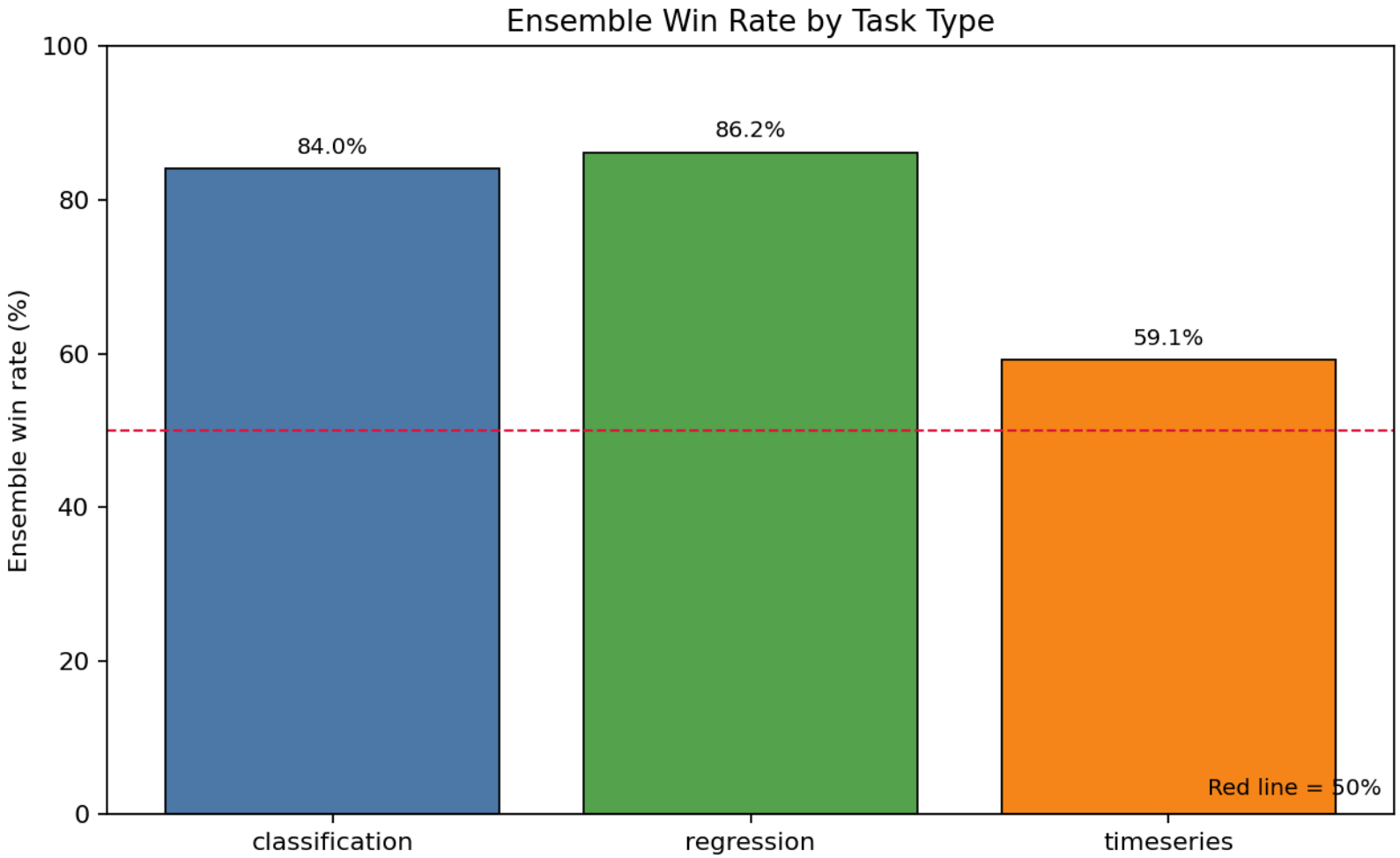
Predictive Model

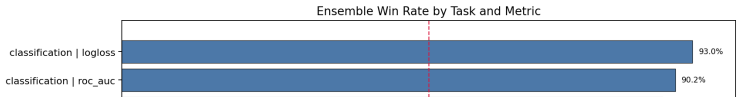


Results

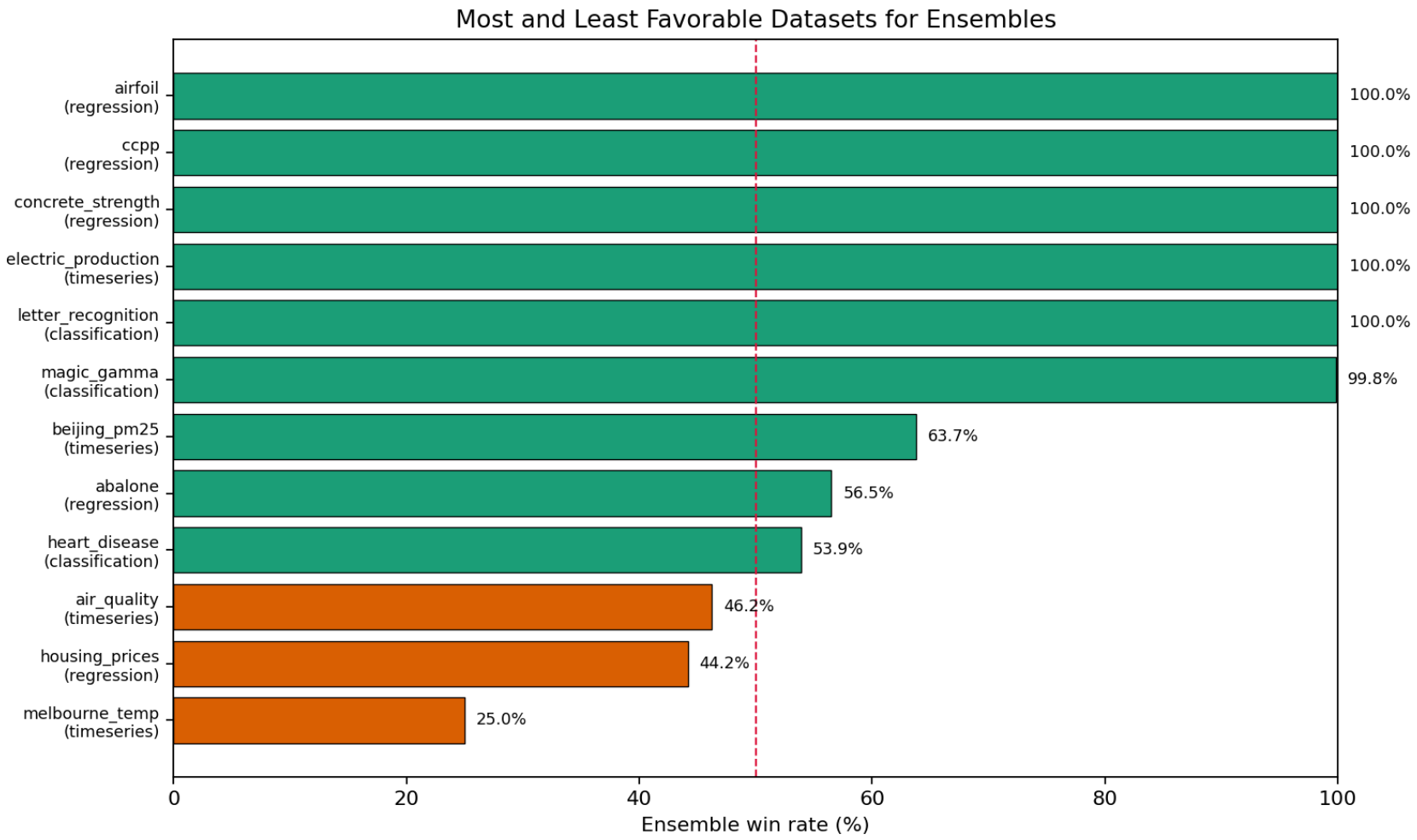

Final message
Regression
Classification
Caveats
Ensembles often strong
Many metrics improve
Results depend on dataset
No universal winner
Warnings matter
Timeouts and failures matter
Metric choice matters

Results

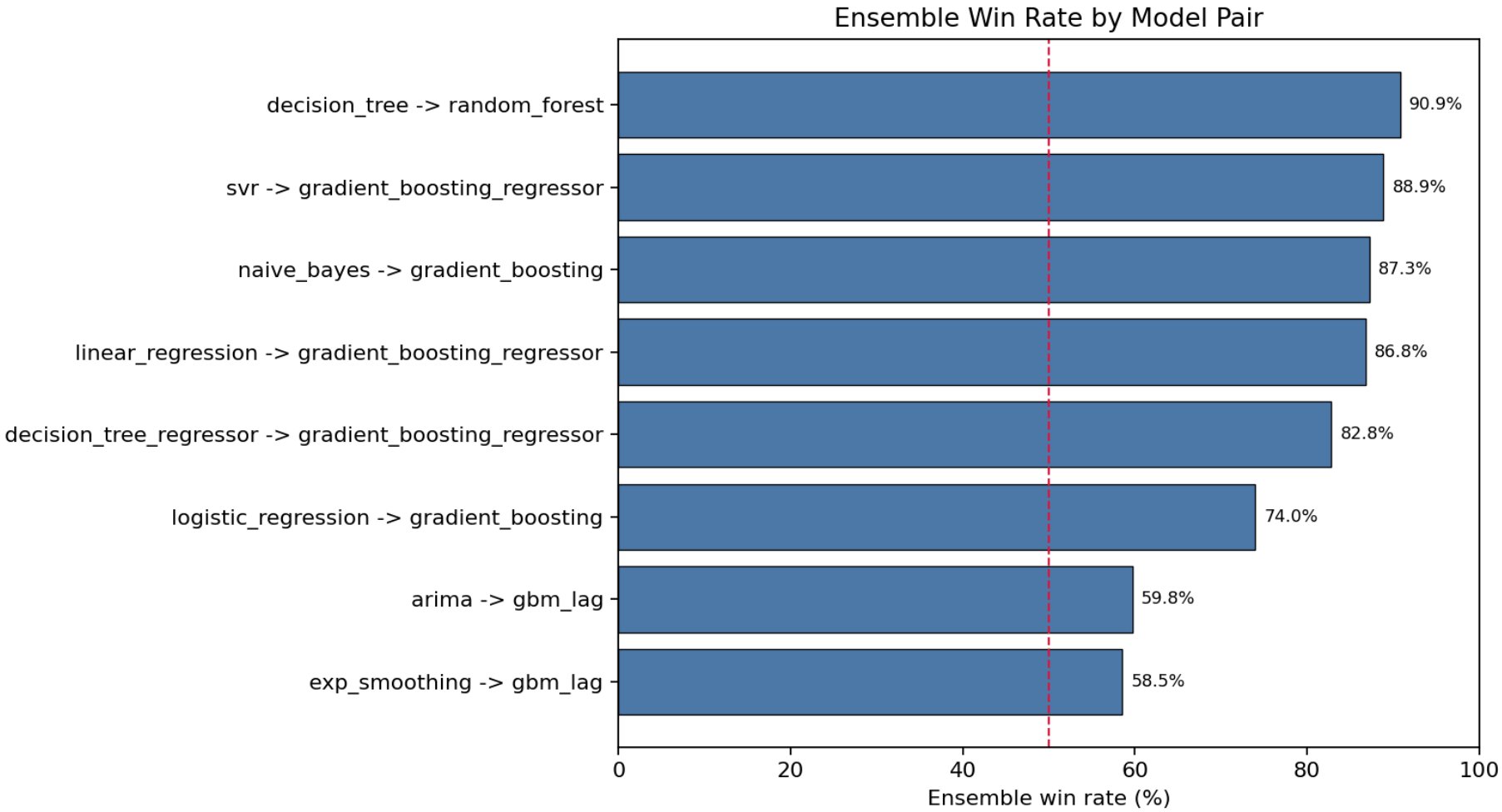

Ensemble Win Rate by Task Type
Ensemble win rate (%)
100
80
60
40
20
0
84.0%
86.2%
59.1%
Red line = 50%
classification
regression
timeseries


Ensemble Win Rate by Task and Metric
classification | logloss
classification | roc_auc
93.0%
90.2%

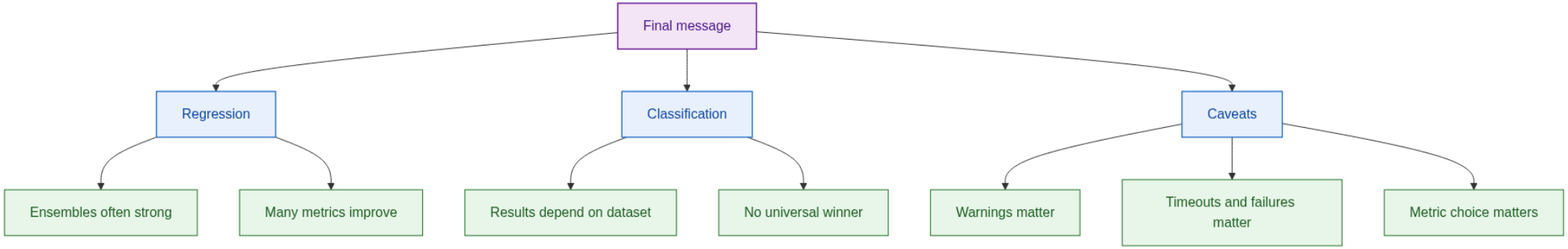
Results


Most and Least Favorable Datasets for Ensembles
airfoil (regression)
ccpp (regression)
concrete_strength (regression)
electric_production (timeseries)
letter_recognition (classification)
magic_gamma (classification)
beijing_pm25 (timeseries)
abalone (regression)
heart_disease (classification)
air_quality (timeseries)
housing_prices (regression)
melbourne_temp (timeseries)
100.0%
100.0%
100.0%
100.0%
100.0%
99.8%
63.7%
56.5%
53.9%
46.2%
44.2%
25.0%
0
20
40
60
80
100
Ensemble win rate (%)

Results

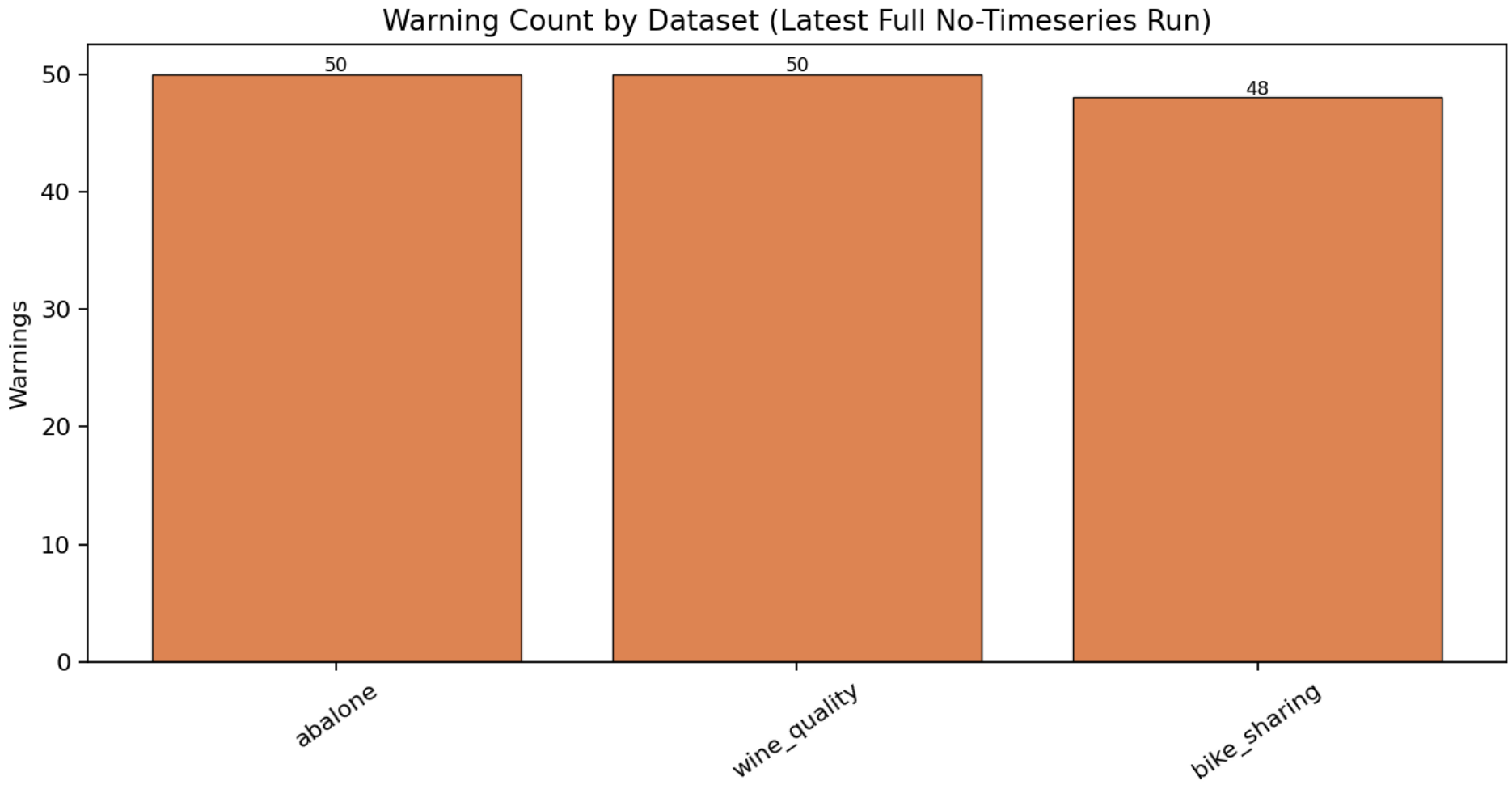

Ensemble Win Rate by Model Pair
decision_tree -> random_forest
svr -> gradient_boosting_regressor
naive_bayes -> gradient_boosting
linear_regression -> gradient_boosting_regressor
decision_tree_regressor -> gradient_boosting_regressor
logistic_regression -> gradient_boosting
arima -> gbm_lag
exp_smoothing -> gbm_lag
90.9%
88.9%
87.3%
86.8%
82.8%
74.0%
59.8%
58.5%
0
20
40
60
80
100
Ensemble win rate (%)

Final Decision



- Ensembles often help, but not always
- Final claim must stay conditional
- Failures, warnings, and mixed datasets matter

Final Decision



Conclusion

- We built fair comparisons across many datasets
- We stored raw results and paired differences
- We analyzed results visually and statistically
- Final conclusion depends on task, dataset, and metric